

Swadesh Swain

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EDUCATION

**GPA conversion to 4.0 scale done according to this website: <https://gpa.eng.uci.edu/node/158>*

Indian Institute of Technology Roorkee, Roorkee, Uttarakhand

GPA: 3.3/4.0*

Bachelor of Technology in Electronics and Communication Engineering

Oct 2022 – Jun 2026

Puna International School, Ahmedabad, Gujarat

GPA: 96.4/100%

Higher Secondary Education

Apr 2020 – Jun 2022

Rachana School, Ahmedabad, Gujarat

GPA: 95.6/100%

Secondary Education

Apr 2007 – Mar 2020

PUBLICATIONS

CroPA++: Exposing Vulnerabilities in Vision Language Models and Enhancing Adversarial Transferability of Cross-Prompt Attacks

NeurIPS 2025, Reliable ML Workshop |  [Open Review](#) |  [PDF](#) |  [Code](#)

Aug 2025

- Introduced CroPA++; three-fold enhancement to the existing Cross Prompt Attack method

Revisiting CroPA: A Reproducibility Study and Enhancements for Cross-Prompt Adversarial Transferability in Vision-Language Models

TMLR 2025 | MLRC 2025 | ★ [Best Paper Award](#) |  [Open Review](#) |  [arXiv](#) |  [Code](#)

June 2025

- Reproducing "An Image is Worth 1000 Lies: Adversarial Transferability Across Prompts on Vision-Language Models", analyzing its limitations in an attempt to further enhance Cross Prompt Attacks for VLMs
- Introduced a novel initialization technique based on providing better attributes to initial noise using Diffusion Models, Improved Cross-Model Transferability and ASR performance

Riemann Sum Optimization for Accurate Integrated Gradients Computation

NeurIPS 2024, Interpretable AI Workshop |  [arXiv](#) |  [Code](#)

Dec 2024

- Developed **RiemannOpt**, a framework to optimize sample point selection for Riemann Sum approximations in Integrated Gradients methods
- Reduced computational costs by up to 4x while maintaining similar performance to baseline methods

High-Throughput Low-Latency Synthetic Data generation by Diffusion

Under Preparation

June 2025

- Devising a resource-efficient data generation pipeline for synthetic data generation of vibrational accelerometer signals of ball bearings
- Introduced a lightweight diffusion architecture achieving 99.7% discriminator confidence against real data, trained with only 100 samples per-class of real-data

PROJECTS AND RESEARCH WORK

IEEE Signal Processing Cup, ICASSP 2025 | *Team Member* |  [Project Report](#)

Sep 2024 – Feb 2025

- A team research project aimed at finding optimal methods for advanced Deepfake detection in images, submitted to the IEEE Signal Processing Cup, ICASSP 2025
- Advisor: **Vinod Pankajakshan**, Associate Professor, Vision and Image Processing Lab – IIT Roorkee

Deep RL-enabled Fortnite Agent

Oct 2024 – May 2024

- A team project ongoing under Data Science Group, to develop an agent to play the highly complex game "Fortnite", using multi-linear ICVF via temporal difference learning and DreamerV3 as world model
- NVIDIA's COSMOS used as the embedding model to embed video streams for training

Reproducing UPop |  [Code](#)

Dec 2023 – Feb 2024

- Reproduced results for paper titled UPop: Unified and Progressive Pruning for Compressing Vision-Language Transformers, ICML 2023, for DeiT model, under the BYOP Reproducibility Track 2024
- Conducted Extensive ablation study and introduced enhancements for optimizations
- Achieved 90% model compression with a mere 5.5% accuracy drop

WORK AND RESEARCH EXPERIENCE

Virginia Tech | *Research Intern*

July 2025 – Present

- Using circuit tracing interpretability methods in modern LLMs to perform guardrailng based on identifiable transcoder features
- Advisor: **Dr. Nagendra Aneja**, Associate Professor, Virginia Tech

AuraML | *Research Intern*

June 2025 – Present

- Implementing text-to-3D scene generation frameworks with Graph Diffusion Models
- Created a text to 3D warehouse generator and editor accessible by an extension through Isaac-Sim for industrial simulation applications
- Advisory Lead: **Arjun Gupta**, CTO - AuraML

Robotics Research Centre, IIIT Hyderabad | *Undergraduate Researcher*

April 2025 – July 2025

- Conducted research to develop generalized skill-based models in Reinforcement Learning using Diffusion Models and Sparse MoE-frameworks for VLA tasks
- Developing Multi-Task RL frameworks
- Advisor: **KM Krishna**, Professor and Lab Head, Robotics Research Centre – IIIT Hyderabad

Koita Centre for Digital Health, IIT Bombay | *Machine Learning Engineer*

May 2024 – Oct 2024

- Implementing pipelines for RAG assisted content generation, OCR correction and content assisted Image reconstruction for medical applications as part of the Digital Health Team
- Project Chair Professor: **Ganesh Ramakrishnan**, Koita Centre for Digital Health – IIT Bombay; Supervisor: **Kundeshwar Pundalik**, Sr. Generative AI Engineer

KonectU | *Generative AI Engineer*

May 2024 – Oct 2024

- Implemented foundational RAG assisted LAM model to autonomously cater a student's application process as part of the early-stage team in this startup, implemented proof of concept models
- Supervisor: **Kundeshwar Pundalik**, , Co-Founder and CTO, KonectU

C.A.N.D.LE Research Lab, IIT Roorkee | **BOSCH** | *Undergraduate Researcher*

Dec 2023 – Present

- Conducting research in collaboration with BOSCH on Geometric manipulation of latent manifolds of Diffusion Models for Editing, Conditioning, and Interpretability
- Synthetic Data Generation using Latent Space Conditioning of Diffusion Models pretrained on fixed length data to arbitrary length data, and Interpretability in Generative Vision Models
- Previous research experience includes in Multi-Modal model compression, and image processing
- Advisor: **Sparsh Mittal**, Associate Professor, C.A.N.D.LE Research Lab – IIT Roorkee

LEADERSHIP & ACTIVITIES

Transactions on Machine Learning Research | TMLR | JMLR | *Reviewer*

Sep 2025 – Present

- Contributing to the reviewing discourse of submitted research works at TMLR

Data Science Group, IIT Roorkee | *Joint Secretary*

Sep 2024 – Present

- Worked on open-source projects and research in the domain of Computer Vision, Diffusion Models, VLMs, Robustness and Explainability
- Managed and organized lectures, workshops, hackathons spanning the field of Data Science and AI
- Our team publishes research at several A* conferences such as NeurIPS, CVPR, ICLR, with most works lead solely by undergraduate authors

International Relations Cell | *Joint Secretary*

Sep 2023 – May 2025

- Created database of research and contacts for various programs and scholarships abroad available for IIT Roorkee students
- Organizing various talk/interview events with partnerships from universities and students from abroad

SELECTED AWARDS AND HONORS

HOD Appreciation Letter for Highest Increase in SGPA in a single semester

Nominated as Institute Representative for NVIDIA AI Summit, Mumbai, India, 2024

All India Rank 1423 out of 1M+ candidates in JEE Advanced 2022

All India Rank 1697 out of 1M+ candidates in JEE Mains 2022
